

Spatial, Vertical, and Asymmetric Onion Price Transmission in India: Evidence from Quantile Nonlinear ARDL and Machine Learning Approaches

Ashwin Deshpande

Data Sciences Lab, DSIM

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Abstract

Onion prices constitute a recurring source of inflationary volatility in India, exerting disproportionate pressure on the Consumer Price Index (CPI) despite a modest commodity weight. This paper develops a system-wide analytical framework to characterise the spatial origins, vertical transmission pathways, asymmetric responses, and temporal persistence of onion price shocks. Utilising daily wholesale arrival data from Agmarknet (2019–2026) and retail price series from the Department of Consumer Affairs (2014–2026), we employ a Quantile Nonlinear Autoregressive Distributed Lag (QN-ARDL) model and a Quantile Random Forest (QRF) augmented with SHAP-based attribution to identify which producer states drive national retail price movements, and for how long. Key findings reveal that Maharashtra and Rajasthan are the dominant price-setters, supply effects persist for up to seven weeks, and consumer states in the eastern and peninsular regions bear the greatest burden. Targeted buffer-stock interventions—especially in Rajasthan—demonstrably reduced inflationary pressure. A Ramsey RESET test confirmed significant non-linearity, motivating the use of quantile and machine-learning methods. The framework provides policymakers operating under India's Flexible Inflation Targeting regime with actionable tools to distinguish transitory shocks from persistent ones.

Index Terms—price transmission, onion, India, NARDL, quantile regression, SHAP, buffer stocks, food inflation, CPI, Koyck distributed lags.

I. INTRODUCTION

Vegetable commodity prices in India represent one of the most volatile segments of the Consumer Price Index (CPI), and onion—colloquially dubbed the "common man's vegetable"—occupies a uniquely prominent position in this volatility. Though Tomato, Onion, and Potato (TOP) collectively carry only a 2.3% weight in the all-India CPI basket (base year 2012=100), they have at various times accounted for nearly a third of headline CPI inflation. In December 2017, 2019, and 2025, the TOP sub-group contributed approximately 34%, 44%, and 45% to headline CPI inflation, respectively, underscoring the outsized macroeconomic significance of these three commodities [1].

Figure 1 plots the national average onion retail price (Rs/kg) from 2014 to 2026 alongside onion's rolling contribution to Food & Beverages and CPI inflation. The price series exhibits a sharp spike-crash pattern, reaching nearly ₹100/kg in 2019 before collapsing. The contribution panel shows that onion periodically drives more than 50% of Food & Beverages inflation, well above its structural weight—dramatising the policy relevance of understanding its transmission dynamics.

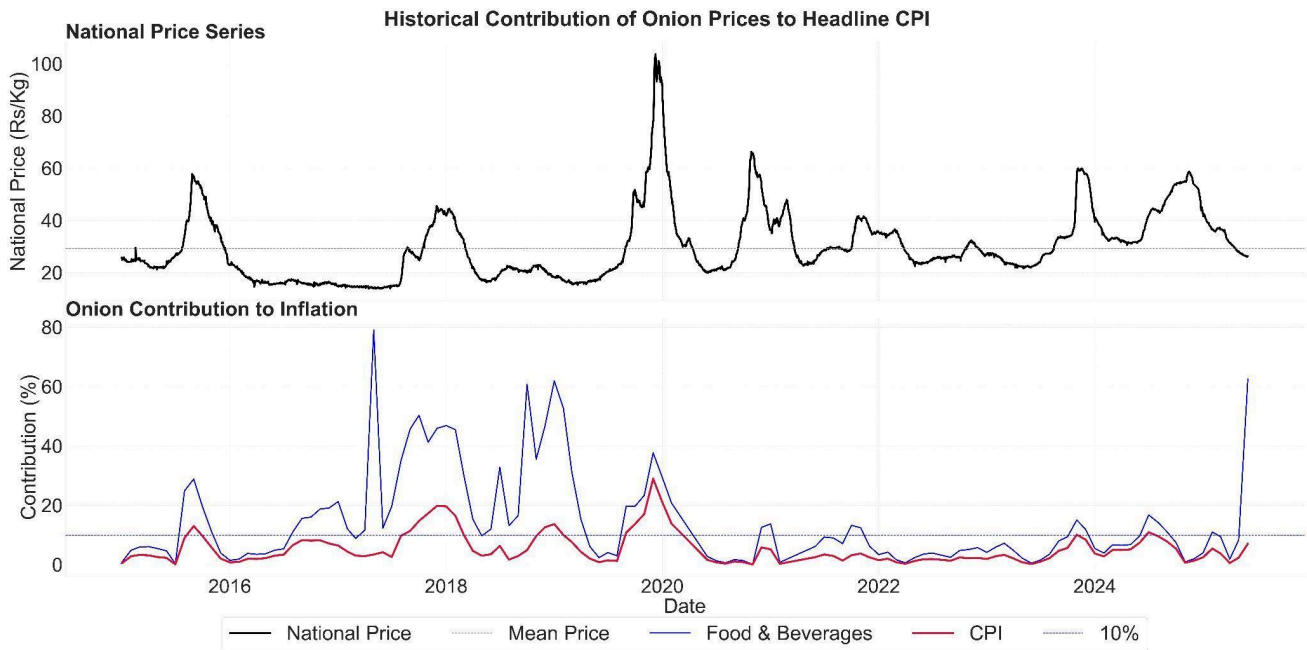


Fig. 1. National onion retail price (Rs/kg, upper panel) and its rolling contribution to Food & Beverages and headline CPI inflation (lower panel), 2014–2026. Dotted lines indicate mean price (upper) and 10% contribution threshold (lower). Source: MoSPI, DoCA.

Price shocks in the onion market are characterised by overlapping causal mechanisms. Supply disruptions attributable to climate variability—unseasonal rainfall, hailstorms affecting major producing belts, and erratic sowing conditions—interact with structural vulnerabilities arising from the highly fragmented nature of Indian agriculture, where most cultivators operate on holdings of two to three hectares. This fragmentation necessitates multi-layer aggregation through mandis (regulated wholesale markets), commission agents, and wholesale traders before produce reaches retail channels. Superimposed on these structural factors are intermittent government interventions: export bans, Minimum Export Prices (MEP), export duties, and buffer-stock operations managed by NAFED, NCCF, and the Department of Consumer Affairs (DoCA) under the Price Stabilisation Fund (PSF). The effect of any given shock may be transitory—rapidly self-correcting once seasonal supply resumes—or persistent, with downstream effects on CPI inflation enduring for months.

This heterogeneity poses a fundamental challenge for monetary policymakers operating under India's Flexible Inflation Targeting (FIT) framework. The Reserve Bank of India targets headline CPI inflation within a band of 4 ± 2 percent; when repeated onion price spikes push the headline index above the upper tolerance limit, policymakers must assess whether the shock is supply-side and transitory—warranting accommodation—or embedded in broader inflation dynamics—warranting rate action. Without precise tools to decompose the magnitude, spatial origin, and duration of onion price shocks, such assessments rely on judgement rather than evidence.

A further dimension of complexity is the spatial character of these shocks. Onion cultivation is concentrated in a small number of states—primarily Maharashtra, Madhya Pradesh, Gujarat, Karnataka, and Rajasthan—yet disruptions in these regions propagate rapidly to consumers across all 28 states and 8 Union Territories. Existing literature on Indian vegetable market integration largely focuses on bilateral market pairs, examines supply-chain efficiency from a producer welfare perspective, or restricts attention to a short historical window [2], [3]. Few studies adopt a system-wide approach capable of simultaneously characterising spatial transmission, asymmetric response to supply increases versus decreases, and the temporal decay of shocks.

This paper addresses these gaps by constructing an integrated analytical framework combining (i) econometric modelling via Quantile Nonlinear ARDL (QN-ARDL) and (ii) machine learning inference via Quantile Random Forest (QRF) with Shapley Additive Explanations (SHAP). The framework is applied to a rich dataset spanning daily wholesale arrivals at the mandi level (2019–2026), daily retail prices at the state level (2014–2026), and monthly CPI sub-indices (2014–2025). Together, these data allow us to estimate supply-price elasticities under different market regimes, identify the most influential producer states, map consumer-state vulnerability, quantify the persistence of supply shocks, and evaluate the efficacy of buffer-stock interventions. The remainder of this paper is organised as follows: Section II presents stylised facts, Section III surveys the literature, Section IV states objectives, Section V describes data, Section VI details the methodology, Section VII presents results, and Section VIII concludes.

II. STYLIZED FACTS

India is both the world's second-largest producer and a major exporter of onions, with annual production exceeding 30 million metric tonnes. Despite this aggregate scale, supply geography is highly concentrated. Five states—Maharashtra (MH), Madhya Pradesh (MP), Gujarat (GJ), Karnataka (KA), and Rajasthan (RJ)—account for the overwhelming share of mandi arrivals. Table I provides average monthly arrival volumes (metric tonnes) across these states, illustrating the seasonal rotation of supply dominance.

Table I. Average Monthly Onion Arrivals by Producing State (Metric Tonnes, 2019–2025)

Season	month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec
Rabi	MH	23,984	26,327	23,724	18,562	22,558	24,810	25,732	23,556	18,982	16,363	14,695	19,861
	MP	2,945	2,801	2,401	4,464	6,063	8,973	6,767	7,065	8,071	7,051	5,949	4,339
Kharif	GJ	7,019	10,060	7,750	6,206	4,310	2,588	1,840	1,521	1,579	1,672	1,977	4,513
	KA	3,256	3,169	3,294	3,089	2,912	2,687	2,355	2,536	3,062	4,718	3,621	3,156
	RJ	965	928	1,194	1,594	1,368	944	876	790	731	750	1,677	1,802

	Total	38,169	43,284	38,258	33,527	36,847	39,618	37,233	35,106	31,987	29,773	27,900	33,671
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Maharashtra dominates supply throughout the year, contributing 53–69% of total mandi arrivals in any given month. Karnataka's share peaks in October (16%), while Gujarat supplies disproportionately in February (23%). Rajasthan, though a smaller contributor in absolute terms (2–6%), is critically important during the lean supply periods of August through October, when Maharashtra and Gujarat supplies are seasonally depressed. Wholesale price data reveals sharp recurring spike-crash cycles, with retail prices breaching ₹80/kg in 2019, 2021, and 2023 before collapsing to below ₹20/kg during peak harvest months. The supply chain structure amplifies price volatility: Lasalgaon in Maharashtra is widely cited as India's premier onion market, though recent evidence suggests that Hubli in Karnataka may serve as price-leader in certain segments [4]. Limited cold-storage infrastructure—less than 3% of output is stored commercially—means that price discovery is concentrated and information asymmetries between farmers and traders remain large.

III. LITERATURE SURVEY

The literature on agricultural price transmission spans several methodological traditions. Classical cointegration approaches using Engle-Granger and Johansen frameworks established that Indian agricultural markets exhibit long-run price co-movement, though short-run deviations can be substantial [5]. Pesaran, Shin, and Smith introduced ARDL Bounds testing, relaxing the requirement that all variables be integrated of the same order [6]. Ibrahim demonstrated that accounting for asymmetry in price adjustment—where the speed of price increase differs from the speed of decrease—substantially improves model fit in commodity markets, motivating the Nonlinear ARDL (NARDL) extension [7]. Roy et al. applied ARDL methods to the TOP commodity complex in India, documenting significant balance-sheet effects and differential transmission [8]. Mohan et al. examined transmission from other commodity subgroups of the CPI to TOP prices, identifying significant cross-commodity spillovers [9].

A foundational concept in modelling the temporal spread of supply shocks is the Koyck distributed lag framework, introduced by Koyck [24]. The Koyck model posits that the impact of a current shock geometrically decays over time—the immediate effect is largest, with each subsequent period's effect being a constant fraction λ ($0 < \lambda < 1$) of the previous period's effect. In the context of onion supply shocks, this implies that a supply disruption in week t affects national prices most acutely in week t , with progressively attenuating effects in weeks $t+1$, $t+2$, and so on. The half-life of a Koyck lag—defined as the number of periods required for the cumulative effect to reach half its long-run total—provides a direct empirical measure of shock persistence. Our results in Section VII-C interpret the SHAP-based lag attribution profiles through this Koyck lens, showing that Rajasthan shocks exhibit a markedly longer half-life than other producer states.

Quantile regression, introduced by Koenker and Bassett [10], enables estimation of covariate effects at multiple conditional quantiles of the outcome distribution, making it particularly suited to analysing price transmission in markets where the relationship may differ between high-price and low-price episodes. Koenker and Xiao extended this to autoregressive settings (QAR) [11], and Chavas applied related quantile autoregressive methods to agricultural markets [12]. In the machine learning literature, Breiman's Random Forest [13] has demonstrated strong predictive performance across economic forecasting. Recent work by Belguith et al. [14] and Jenett et al. [15] combined QRF with SHAP (Shapley Additive Explanations, introduced by Lundberg and Lee [16]) to decompose predictions into interpretable state-level contributions. Madaan et al. [17] documented significant seasonal patterns in onion price formation using Indian mandi data. Paul et al. [18] documented market integration across onion mandis. Paul, Das, and Mathur [4] provide evidence that Hubli, rather than Lasalgaon, exercises price leadership in the contemporary Indian onion market.

IV. RESEARCH OBJECTIVES

This study pursues four interconnected objectives:

- O1. Spatial origin of price shocks:** Identify which producer states drive onion price changes nationally and quantify state-specific supply-price elasticities across different price regimes.
- O2. Temporal persistence:** Measure how many weeks a supply shock in a given producer state continues to exert statistically significant influence on national retail prices, interpreted via the Koyck half-life metric.
- O3. Spatial vulnerability of consumer states:** Map the differential transmission of producer-region price shocks across Indian states, identifying which consumer populations bear the greatest burden.
- O4. Policy efficacy:** Evaluate the effect of government buffer-stock releases and procurement operations on national and state-level retail price trajectories.

V. DATA

A. Agmarknet Wholesale Data

Daily wholesale prices and arrival quantities (metric tonnes) at the individual mandi level for onion are obtained from the Agmarknet portal, maintained by the Directorate of Marketing and Inspection, Ministry of Agriculture and Farmers Welfare. The dataset spans January 2019 to April 2026, covering mandis across all major onion producing and trading states. These data enable construction of state-level aggregate arrival indices used in all price transmission models.

B. Department of Consumer Affairs (DoCA) Price Data

Daily retail and wholesale prices at the state level are sourced from the DoCA price monitoring portal, covering January 2014 through April 2026. The DoCA series is used both as the dependent variable in state-level price transmission models and as a validation dataset for model-generated forecasts.

C. MoSPI CPI Sub-Indices

Monthly CPI indices at the commodity, group, and headline level—disaggregated by state and for All-India—are obtained from the Ministry of Statistics and Programme Implementation. The series covers January 2014 through December 2025. The vegetable sub-index and the combined Food and Beverages group are used to contextualise onion price dynamics within the broader inflation framework.

VI. METHODOLOGY

A. Preliminary Diagnostics

Before specifying the main models, we conduct three diagnostic tests. First, unit root tests—Augmented Dickey-Fuller (ADF) [19] and Phillips-Perron (PP) [20]—are applied to all price and arrival series to determine integration orders. Second, we test for quantile heterogeneity via pairwise Wald tests and a Global Constancy Test comparing coefficients at quantiles $\tau \in \{0.1, 0.25, 0.5, 0.75, 0.9\}$. Third, we apply the Ramsey RESET test [21] for functional form misspecification. A restricted linear model is first estimated:

$$\Delta Y_{t+1} = \alpha + \sum_i [\varphi_i \cdot \Delta Y_{t-i}] + \sum_r (\beta_{r,i} \cdot \Delta X_{r,t-i}) + \varepsilon_t$$

An unrestricted model augments this with squared and cubed fitted values from the restricted model:

$$\Delta Y_{t+1} = (\text{restricted model}) + \gamma_1 \cdot \Delta \hat{Y}_{t+1}^2 + \gamma_2 \cdot \Delta \hat{Y}_{t+1}^3 + \varepsilon_t$$

The null $H_0: \gamma_1 = \gamma_2 = 0$ is tested via an F-statistic. Our results yield $F = 3.52$ ($p = 0.031$), firmly rejecting linearity. LOESS smoothing [23] is applied throughout to extract underlying trends from noisy arrival and price series.

B. Koyck Distributed Lags and Shock Persistence

A central theoretical framework for modelling the temporal spread of supply shocks is the Koyck distributed lag model [24]. The core Koyck specification posits that each additional lag of an explanatory variable exerts a geometrically declining effect on the outcome:

$$y_t = \alpha + \beta_0 \cdot x_t + \beta_0 \lambda \cdot x_{t-1} + \beta_0 \lambda^2 \cdot x_{t-2} + \dots + \varepsilon_t, \quad 0 < \lambda < 1$$

The parameter λ is the geometric decay (retention) rate: the fraction of each week's supply-shock effect that carries over to the next week. The half-life of the shock—the number of weeks k such that the cumulative effect at k is half the long-run total—is given by $k = \log(0.5) / \log(\lambda)$. A λ close to 1 implies long persistence; λ close to 0 implies rapid decay. We use the Koyck half-life as the interpretive lens for comparing the SHAP-derived lag profiles across states (Section VII-C).

C. Model 1: Quantile Nonlinear ARDL (QN-ARDL)

To account simultaneously for asymmetry and quantile heterogeneity, we extend the NARDL framework of Ibrahim [7] to a quantile setting. Positive and negative partial sum decompositions of each regional arrival variable are constructed:

$$\begin{aligned} x^+_{r,t} &= \sum_{i=1}^t \max(\Delta x_{r,i}, 0) \\ x^-_{r,t} &= \sum_{i=1}^t \min(\Delta x_{r,i}, 0) \end{aligned}$$

The QN-ARDL model for quantile τ is then:

$$\begin{aligned} Q_{\{y_{t+1}\}}(\tau | F_t) &= \alpha(\tau) + \beta_0(\tau) \cdot y_t + \sum_r [\beta^+_r(\tau) \cdot x^+_{r,t} + \beta^-_r(\tau) \cdot x^-_{r,t}] \\ &+ \sum_i \rho_i(\tau) \cdot \Delta y_{t-i} + \sum_i \sum_r [\gamma^+_{r,i}(\tau) \cdot \Delta x^+_{r,t-i} + \gamma^-_{r,i}(\tau) \cdot \Delta x^-_{r,t-i}] + \varepsilon_t \end{aligned}$$

Long-run asymmetric multipliers are $\gamma^+_r(\tau) = \sum_i \gamma^+_{r,i}(\tau)$ and $\gamma^-_r(\tau) = \sum_i \gamma^-_{r,i}(\tau)$. Cointegration is verified using ARDL Bounds Testing [6]; the calculated F-statistic of 6.83 ($k = 4$ regressors) exceeds the Pesaran et al. upper critical bound at 1%. An asymmetry Wald test is conducted for each state r at each quantile τ to determine whether $\beta^+_r(\tau) \neq \beta^-_r(\tau)$. Lag order p is selected by Schwarz Information Criterion (SIC) [22], which selects $p = 3$.

D. Model 2: Quantile Random Forest with SHAP Attribution

As a complement to the parametric QN-ARDL approach, we estimate a Quantile Random Forest (QRF) [13] applied to the same feature set of lagged national prices and lagged positive and negative partial sums of regional arrivals. QRF grows an

ensemble of 500 decision trees and estimates conditional quantile functions by taking the empirical quantile of training observations in the leaf node mapped to each test observation. The QRF's predictions are decomposed using SHAP values [16]:

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} [|S|! (|F| - |S| - 1)! / |F|!] \cdot [f_{S \cup \{j\}}(x_{S \cup \{j\}}) - f_S(x_S)]$$

By aggregating SHAP values over lag indices and across the price distribution, we derive state-specific temporal contribution profiles—identifying not only which states drive prices but for how many weeks back their supply remains influential. The decay profile of these lagged SHAP attributions is then interpreted through the Koyck lens (Section VI-B): a faster-decaying profile implies a shorter half-life and lower λ .

VII. RESULTS

A. Model Specification Tests

Unit root tests confirm that all price series are $I(1)$ while arrival quantities are $I(0)$, validating the ARDL bounds approach. The Ramsey RESET test yields $F = 3.52$ ($p = 0.031$), rejecting the null of correct linear specification. Pairwise Wald tests across quantiles reject coefficient constancy at all state-quantile combinations, confirming that supply-price response is statistically distinct across the price distribution. The ARDL Bounds F-statistic of 6.83 exceeds the Pesaran et al. upper critical bound at the 1% level, establishing long-run cointegration.

B. Supply-Price Elasticities by State and Price Regime

Figure 2 provides an arrow-plot visualisation of the QN-ARDL long-run supply-price elasticities across all states at three quantile regimes. The green upper panel shows responses to a +10% supply increase; the pink lower panel shows responses to a -10% supply decrease. The scale of effects at the High Price (Low Supply) regime (rightmost column) is strikingly larger than at the other two regimes, visually confirming the strong non-linearity. Each coloured arrow represents one state: Gujarat (gold), Karnataka (dark red), Madhya Pradesh (teal), Maharashtra (blue), Rajasthan (magenta), and the aggregate Total (black).

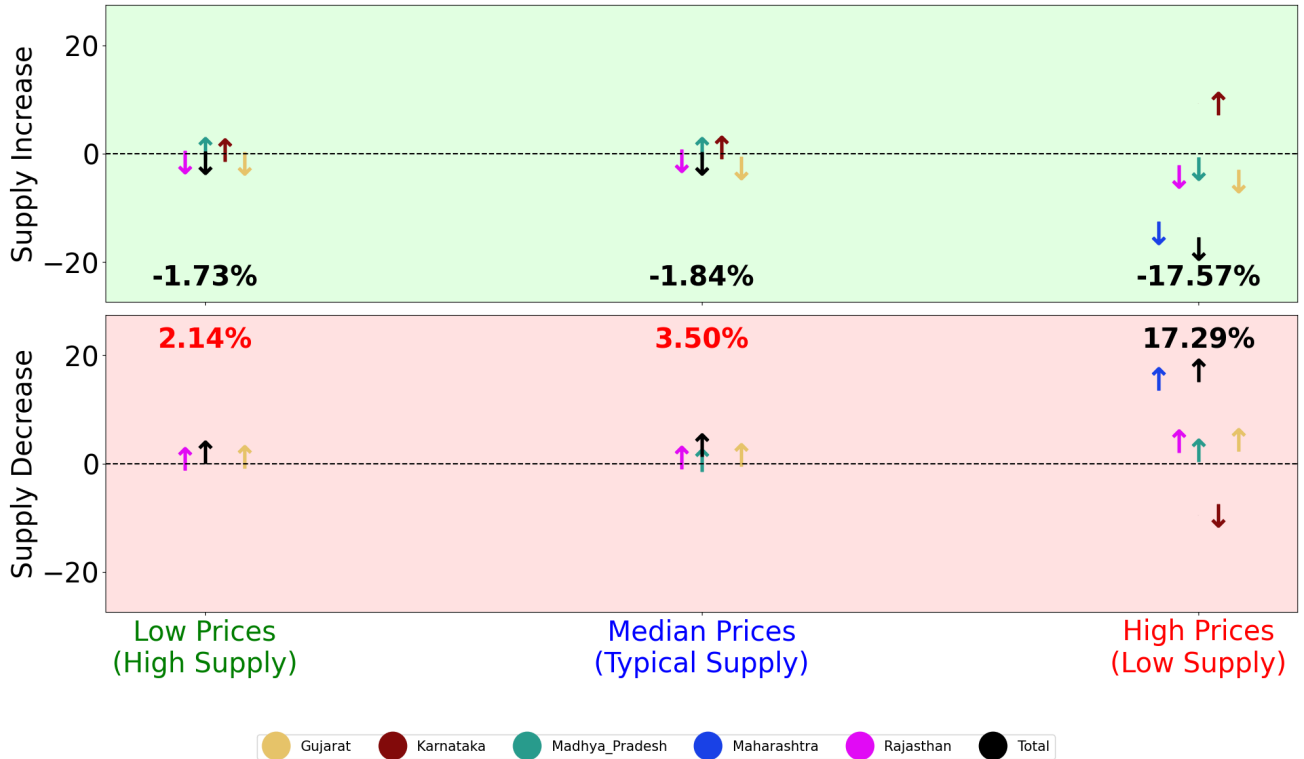


Fig. 2. QN-ARDL long-run supply-price elasticities (%) by state and price regime. Upper panel (green): response to a +10% supply increase; lower panel (pink): response to a -10% supply decrease. Arrow direction indicates price response direction; aggregate net effects shown in bold. Source: Authors' estimates.

Table II presents the numerical elasticities from the QN-ARDL model. Maharashtra exhibits strong asymmetric effects at high price regimes: a 10% supply increase during a high-price episode is associated with a -14.70% reduction in national prices, while a 10% supply reduction pushes prices up by +15.71%. This rockets-and-feathers asymmetry means that price spikes propagate more forcefully than equivalent supply increases dampen them. The aggregated All-States result confirms

strong non-linearity: the high-price regime exhibits elasticities ten times larger than the low-price regime for supply decreases.

Table II. QN-ARDL Long-Run Supply-Price Elasticities (%) by State and Price Regime

State	Change in Supply	Low Prices ($\tau < 0.25$)	Median Prices ($0.25 < \tau < 0.75$)	High Prices ($\tau > 0.75$)
Maharashtra	+10%			-14.70
	-10%			15.71
Madhya Pradesh	+10%	0.99	0.99	-2.76
	-10%		0.71	2.54
Gujarat	+10%	-1.80	-2.62	-5.12
	-10%	1.24	1.64	4.42
Karnataka	+10%	0.69	1.20	9.27
	-10%			-9.59
Rajasthan	+10%	-1.61	-1.41	-4.25
	-10%	0.90	1.15	4.22
Total	+10%	-1.73	-1.84	-17.57
	-10%	2.14	3.50	17.29

* Net effect (combined asymmetric elasticity at high price regime)

C. Temporal Persistence of Supply Shocks (Koyck Half-Life Analysis)

Figure 3 presents the SHAP-derived lag decay curves for each state across the three price regimes, plotted up to 20 weeks. These curves represent the estimated weekly price-growth contribution (%/week) attributable to each state's supply, as a function of lag length. Interpreted through the Koyck framework, the curvature of each line encodes the state's λ parameter: a steeper initial decline followed by near-zero attribution implies a short half-life, while a flatter, more gradually declining profile implies a longer half-life and greater persistence.

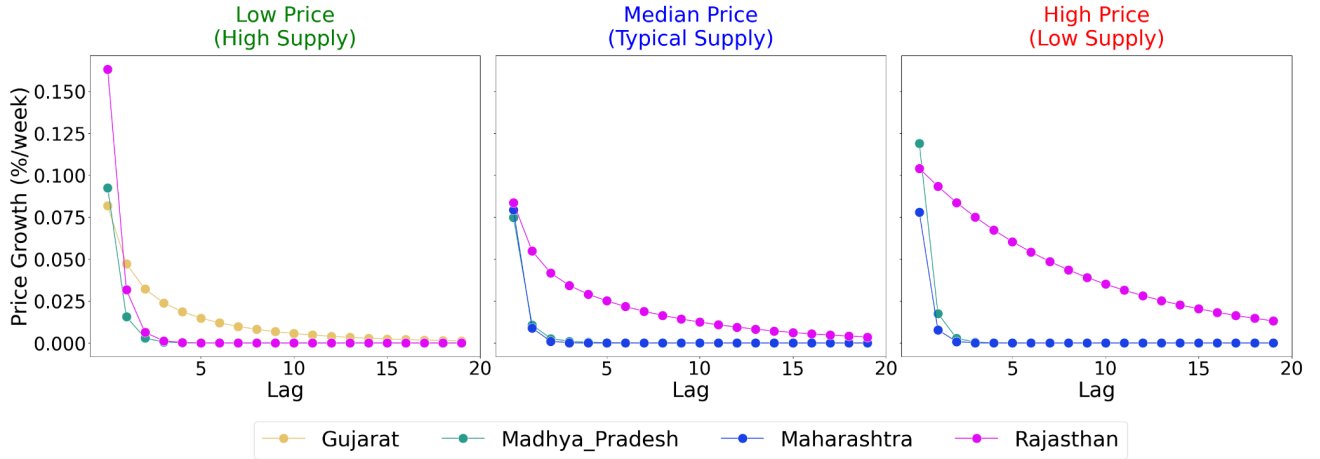


Fig. 3. Koyck-style lag decay profiles of supply-shock contributions (%/week) by state and price regime. Horizontal axis: lag in weeks (1–20). Vertical axis: estimated price-growth contribution per unit supply shock. Rajasthan (magenta) exhibits the longest Koyck half-life, especially under the High Price regime. Source: QRF-SHAP estimates.

Figure 4 quantifies these half-lives directly using boxplots of Koyck half-life estimates (in weeks) derived from fitting the geometric decay model to each state's SHAP lag profile, across price regimes. The results reveal dramatic heterogeneity. Rajasthan's half-life during the High Price regime (red box) spans a wide range extending to 10+ weeks, reflecting high uncertainty about the exact decay rate but consistently implying the longest persistence across all states. Maharashtra's half-life clusters tightly below 1 week across all regimes—its shocks are large but decay rapidly. Madhya Pradesh and Gujarat both exhibit compact half-life distributions around 0.3–0.5 weeks, meaning that their supply effects are transmitted almost entirely within the first two weeks. Gujarat's Low Price regime shows a somewhat longer half-life, consistent with its role as a swing supplier whose lean-season shortfalls create stickier price effects.

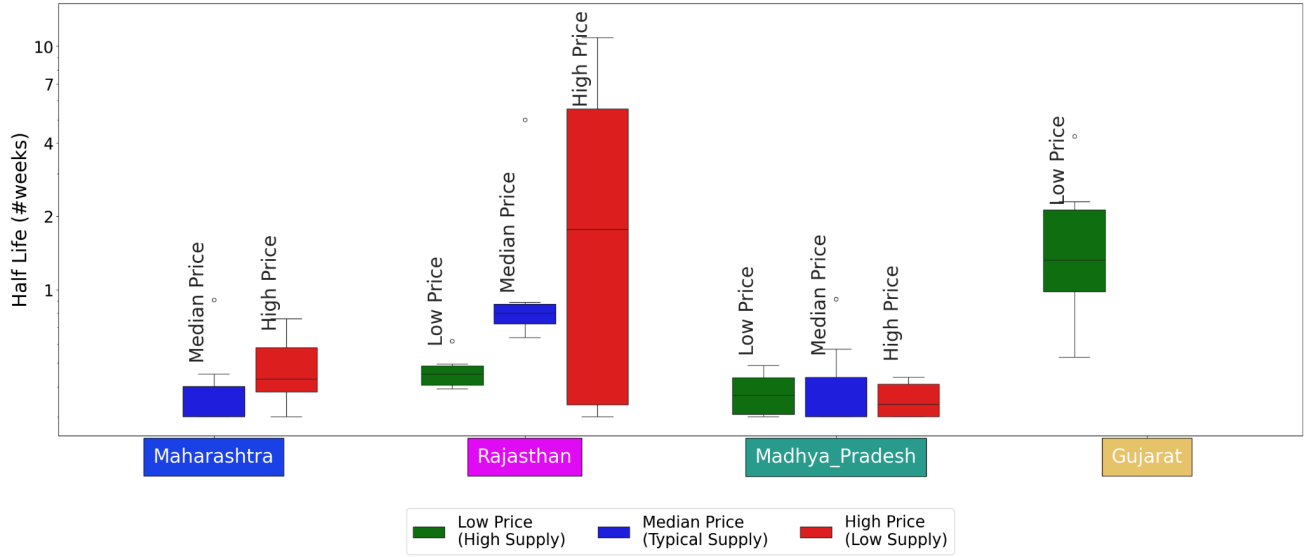


Fig. 4. Koyck half-life estimates (weeks) by state and price regime, derived from fitting geometric decay to QRF-SHAP lag attribution profiles. Log-scale vertical axis. Blue = Low Price (High Supply); Green = Median Price; Red = High Price (Low Supply). Rajasthan exhibits the widest and highest half-life distribution under high-price regimes, confirming the longest shock persistence. Source: Authors' estimates.

The practical implication of Rajasthan's long Koyck half-life is significant: a supply shock from Rajasthan during a lean period (August–October) will still be meaningfully influencing national prices 6–7 weeks later. This exceeds the typical policy response lag—meaning that early identification of Rajasthan supply shortfalls creates a viable intervention window. Maharashtra's short half-life, conversely, implies that even large supply-side disturbances in Maharashtra are largely self-correcting within 2–3 weeks, reducing the urgency for monetary policy response to purely Maharashtra-driven spikes.

D. Spatial Vulnerability of Consumer States

Figure 5 presents the all-India price shock transmission choropleth map. The colour intensity (light pink to deep dark red) encodes the estimated impact magnitude of a combined producer-state supply shock on each consumer state's retail price. The geographic pattern is strikingly non-uniform: eastern states (Uttar Pradesh at 18.44%, West Bengal at 14.67%), the peninsular state of Andhra Pradesh (22.31%), and the landlocked Chhattisgarh (17.10%) are most heavily affected. By contrast, northern and western states proximate to the major producing belts show lower impact magnitudes.

All India Price Shock Transmission

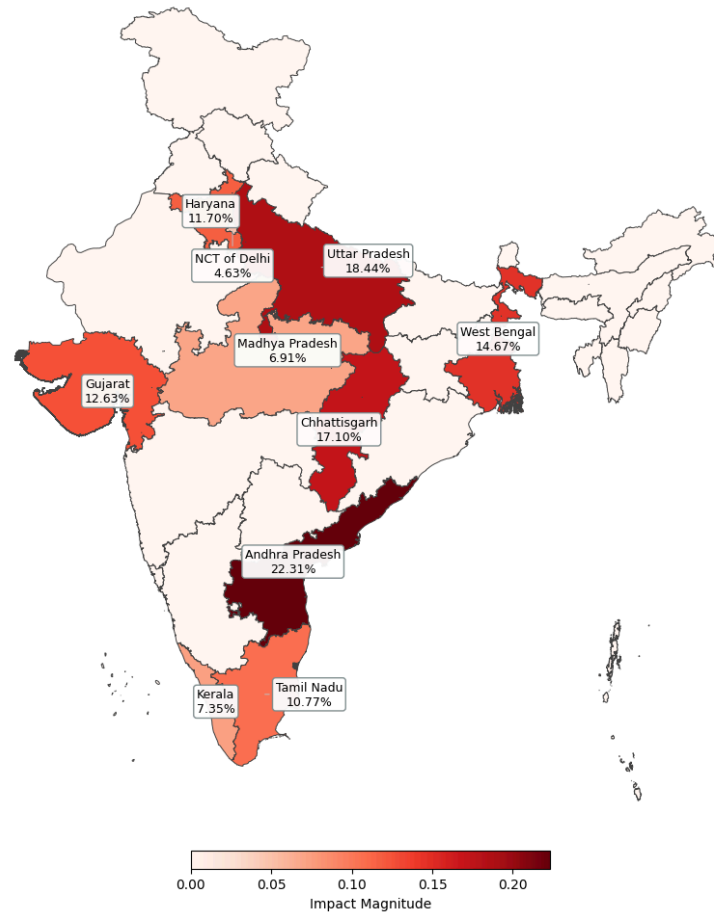


Fig. 5. All-India price shock transmission map. Colour intensity (light pink to dark red) encodes impact magnitude of a combined producer-state supply shock on each state's retail onion price. Andhra Pradesh (22.31%), Uttar Pradesh (18.44%), Chhattisgarh (17.10%), and West Bengal (14.67%) are the most exposed consumer states. Source: QN-ARDL estimates.

Figures 6 and 7 decompose this aggregate map into the individual transmission fingerprints of Maharashtra and Rajasthan, respectively—the two dominant price-setting states. Maharashtra's transmission (Fig. 6) is strongest to Uttar Pradesh (12.03%), Kerala (8.76%), Chhattisgarh (7.30%), and West Bengal (6.81%). The flow arrows indicate the primary supply-chain routing through which Maharashtra price shocks reach these distant consumers. Rajasthan's transmission (Fig. 7), by contrast, is most intensely felt in Bihar (10.21%), Andhra Pradesh (7.73%), and Chhattisgarh (4.21%), reflecting Rajasthan's critical role as a lean-season marginal supplier to deficit regions.

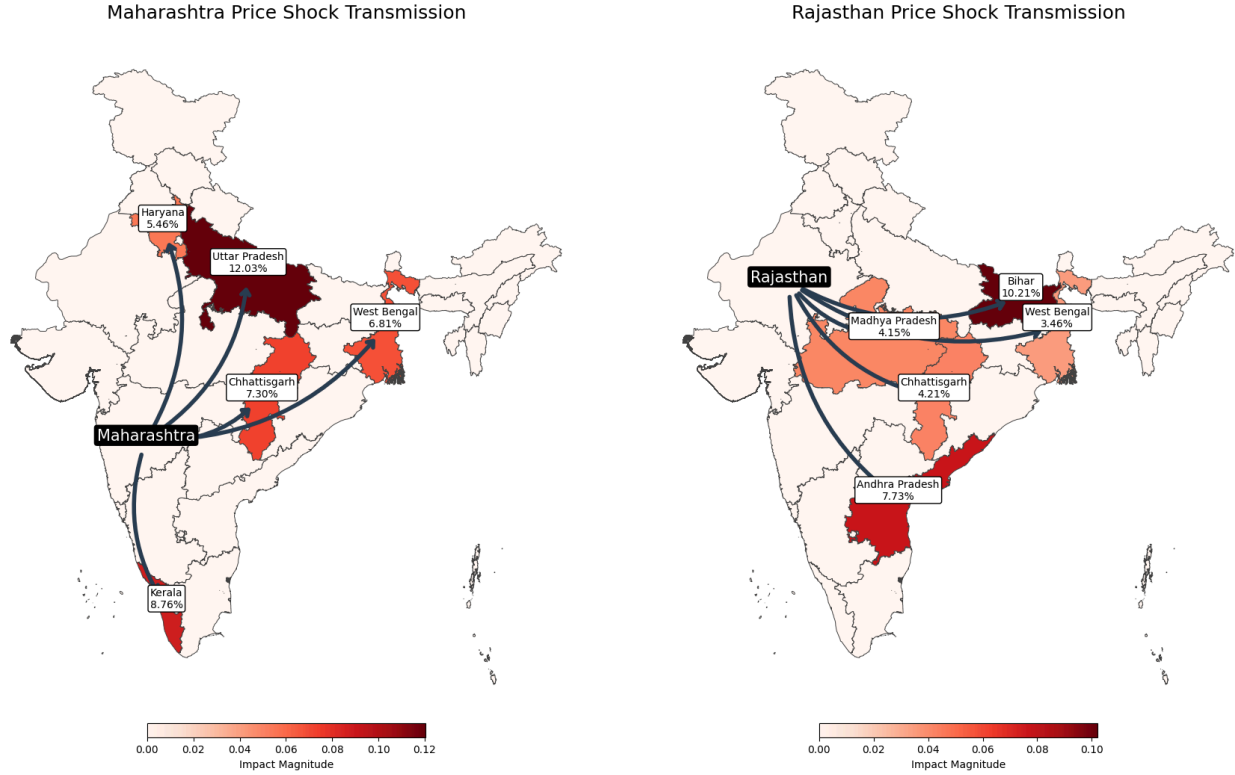


Fig. 6 (left). Maharashtra price shock transmission map. Arrows indicate primary flow corridors; intensity encodes impact magnitude. Fig. 7 (right). Rajasthan price shock transmission map. Rajasthan shocks flow primarily to eastern and peninsular deficit states, with Bihar (10.21%) the most impacted. Source: QN-ARDL estimates.

E. Buffer Stock Intervention Efficacy

Figure 8 presents SHAP value decompositions by region and period (Before Peak, Normal, After Peak) for the four major states, evaluated around the documented government buffer-stock intervention episodes in November 2021, 2023, and 2024. Before buffer-stock activation, Rajasthan's supply attribution (pink bars, Before Peak) was contributing a SHAP value equivalent to +₹22.73/kg of upward price pressure. After buffer-stock releases from PSF reserves, this attribution reversed sharply to -₹7.33/kg (pink After Peak bar, now negative). The net swing of approximately ₹30/kg is economically substantial.

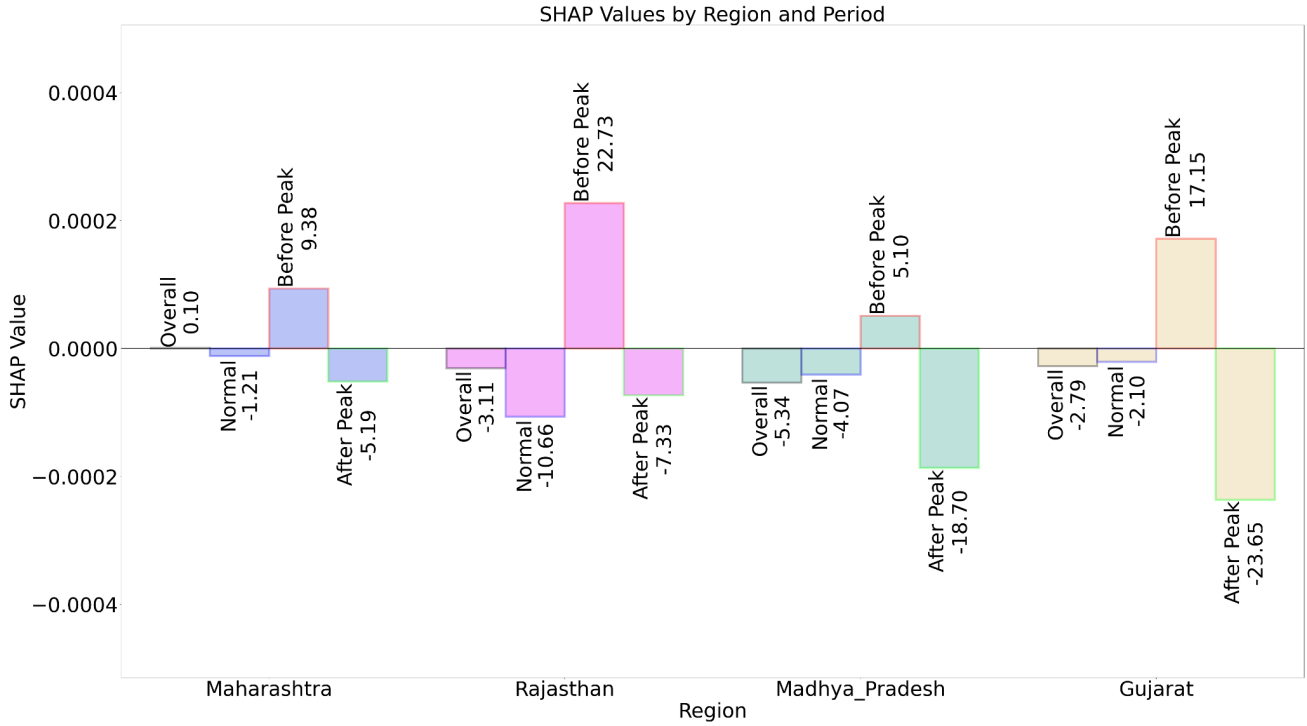


Fig. 8. SHAP value decomposition by region and intervention period (Before Peak, Normal, After Peak). Positive SHAP values (upward-pointing bars) indicate price-increasing supply pressure; negative values indicate price-reducing effect. Rajasthan's Before Peak bar (+22.73) reverses to After Peak (−7.33) following buffer-stock releases, demonstrating a net ₹30/kg policy impact. Source: QRF-SHAP estimates.

Gujarat's After Peak contribution (−23.65) is the largest single negative SHAP value in the chart, reflecting the sustained price-dampening effect of the Rabi season harvest flowing through Gujarat mandis into national markets. Maharashtra's After Peak value (−5.19) is more modest, consistent with its shorter Koyck half-life—Maharashtra supply normalisation has a large but fast-decaying effect. Madhya Pradesh After Peak (−18.70) reflects the cumulative effect of MP's lean-to-peak supply transition, which acts as a sustained buffer over a longer horizon than Maharashtra. These results collectively confirm that targeted, timely supply interventions—especially Rajasthan-sourced buffer releases deployed within the Koyck persistence window—are effective tools for price stabilisation.

VIII. CONCLUSION

This paper has developed and applied a system-wide analytical framework for understanding onion price transmission in India, combining Quantile Nonlinear ARDL econometrics with Quantile Random Forest machine learning and SHAP-based attribution, interpreted through the Koyck distributed lag framework. The framework jointly characterises spatial origin, asymmetric response, temporal persistence, and consumer vulnerability in a manner that prior single-market and linear studies were unable to achieve.

Our principal findings are as follows. (1) Maharashtra and Rajasthan are the dominant price-setters nationally, with Maharashtra contributing the largest magnitude shocks and Rajasthan exhibiting the longest Koyck half-life of up to seven weeks. (2) Supply-price relationships are strongly asymmetric: supply decreases drive price increases far more forcefully than equivalent supply increases moderate them, with high-price regimes exhibiting elasticities ten times larger than low-price regimes. (3) Consumer states in the eastern and peninsular regions—particularly Andhra Pradesh (22.31%), Uttar Pradesh (18.44%), Chhattisgarh (17.10%), and West Bengal (14.67%)—are most exposed. (4) Government buffer-stock interventions, particularly Rajasthan-sourced releases, demonstrably reduced inflationary pressure by an estimated ₹30/kg net swing during documented spike episodes.

For monetary policymakers under India's Flexible Inflation Targeting framework, these results offer practical shock characterisation tools. A supply disruption originating in Rajasthan during a high-price regime carries a qualitatively different signature than a Maharashtra shock: longer Koyck half-life, broader geographic diffusion (especially eastward), and a more favourable intervention window. Distinguishing these dimensions—rather than reacting uniformly to headline CPI vegetable inflation—can reduce the risk of unnecessary policy tightening in response to supply-side shocks that are inherently transitory. For market intervention design, the Koyck framework implies that effective buffer-stock deployment must occur within the first two weeks of a Rajasthan shock to intercept peak transmission. The differential vulnerability profile across consumer states suggests that spatially targeted relief would improve cost-effectiveness over uniform national releases.

Future work should extend this framework to Tomato and Potato, incorporate real-time monitoring for early-warning applications, and explore the interaction between export policy changes and domestic price transmission dynamics.

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